

Conditional branching: if, '?'

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Sometimes we need to perform different actions based on different conditions. To do that we can use the `if` statement and the conditional operator `'?'` That's also called a question mark, operator.

The 'if' statement

The `if(...)` statement evaluates a condition in parentheses and, if the result is true, executes a block of code.

For example:

How to take data input in NodeJS?

1. `npm install prompt-sync [install-prompt sync]`
2. Go to the code editor:

```
const prompt = require('prompt-sync')();  
//data input using prompt()  
let year = prompt('Enter year: ');  
console.log(`Year: ${year}`);
```

← Compulsory in every program, where you are taking data input from the user.

```
const prompt = require('prompt-sync')();  
let year = prompt('Enter year: ');  
console.log(`Year: ${year}`);  
console.log(typeof(year));  
let y = Number(year);  
if (y % 400 == 0 || (y % 100 != 0 && y % 4 == 0)) {  
    console.log(`${y} is a leap year`);  
}  
else {  
    console.log(`${y} is not a leap year`);  
}
```

Century year
1900 ← Not leap year
2000 ←
2100 ←

OR { 1st condition is true, then it will not check the 2nd condition }
1st condition is false, then it will check the 2nd condition }

Boolean conversion

The `if(...)` statement evaluates the expression in its parenthesis and converts the result to a boolean.

Let's recall the conversion rules:

- A number `0`, an empty string `""`, `null`, `undefined`, `NaN` all become false. Because of that they are called "falsy" values.
- Other values become true, so they are called "truthy".

So the code under the condition would never execute:

```
if(0){  
    console.log(`This will not execute!`)  
}  
else{
```

```
    console.log(`This is else,...`)  
  }
```

...and inside this condition-it always will:

```
if(1){  
  console.log(`This will execute!`)  
}  
else{  
  console.log(`This is else,...`)  
}
```

We can also pass a pre evaluated boolean value to if like this:

```
let cond = (y == 2026 ); // Equality evaluates to true or false  
if(cond){  
  console.log(`the year is 2026`);  
}  
else{  
  console.log('Other year');  
}
```

The 'else' clause

The if a statement may contain an optional else block. It executes when the condition is falsy.

Several conditions: "else if"

Sometimes we do like to test several variants of a condition. The else if clause lets us do that:

```
if(y < 2026){  
  console.log(`Too early...`);  
}  
else if(y > 2026){  
  console.log(`Too late`);  
}  
else{  
  console.log(`Exactly!`);  
}
```

Conditional operator '?'

Sometimes we need to assign a variable depending on a condition.

The so called "conditional" or "?" operator lets us do that in a shorter and simpler way.

The operator is represented by "?" Sometimes it is called ternary because the operator has three operands. It is actually one and only operator in Javascript which has that many.

The syntax is:

```
let result = condition ? Value1 : value2;
```

The condition is evaluated: If it's truthy, then value1 is returned, otherwise value2.

```
let age = Number(prompt('Enter age: '))
let accessAllowed = (age > 18) ? true : false;
if(accessAllowed){
    console.log(`${age} is assessible`)
}
else{
    console.log(`${age} is not assessible`)
}
```

Multiple '?'

A sequence of question mark operator '?' Can the return of value that depends on more than 1 condition.

```
let message = (age < 3) ? 'Hi, baby!' :
    (age < 18) ? 'Hello!' :
    (age < 100) ? 'Greetings!' :
    'What an unusual age!';
console.log(message)
```

Homework Assignment: JavaScript - Conditional Branching (if, else, else if, and Ternary ? Operator)

Subject: JavaScript Programming (Node.js)

Topic: Conditional Branching (if, else, else if, ?), Boolean Conversion, User Input using prompt-sync

Difficulty Level: Medium

Instructions:

- Use **Node.js** and **prompt-sync** for user input in **every program**.
- Perform proper type conversion using **Number()** wherever numerical input is required.
- Display meaningful messages for every output.
- Validate user input whenever necessary.

1. Leap Year Checker with Explanation

Write a JavaScript program that accepts a year from the user and determines whether it is a leap year or not. The program must correctly handle century years (e.g., 1900, 2000, 2100) using the leap year rules. Display an appropriate explanation along with the result.

2. Student Grade Evaluation System

Write a JavaScript program that accepts marks (0-100) from the user and displays the corresponding grade using an **if-else-if ladder**.

Use the grading scheme:

- 90-100 → Grade A+
- 80-89 → Grade A
- 70-79 → Grade B
- 60-69 → Grade C
- 50-59 → Grade D
- Below 50 → Fail

Also display whether the student is eligible for scholarship (marks ≥ 85).

3. Voting Eligibility and Citizenship Verification

Write a JavaScript program that accepts the user's age and citizenship status ("Indian" or "Non-Indian").

The program should:

- Check whether the user is eligible to vote.
- Display different messages for:
 - Eligible voter
 - Underage citizen
 - Non-citizen
- Use nested `if` statements.

4. Electricity Bill Calculator

Write a JavaScript program that accepts the number of electricity units consumed and calculates the bill according to the following slabs:

- First 100 units $\rightarrow$ ₹5/unit
- Next 100 units $\rightarrow$ ₹7/unit
- Remaining units $\rightarrow$ ₹10/unit

Display the total bill and the billing category.

5. Largest Among Three Numbers

Write a JavaScript program to input three numbers from the user and determine the largest number using nested `if-else` statements.

Also display whether any two or all three numbers are equal.

6. Simple Calculator using `if-else`

Write a menu-driven calculator that accepts:

- First number
- Second number
- Operator (+, -, *, /, %)

Perform the appropriate arithmetic operation using `if-else-if` statements.

Handle division by zero appropriately.

7. Login Authentication System

Write a JavaScript program that accepts:

- Username
- Password

Use nested `if` statements to verify the credentials.

Conditions:

- Username: admin
- Password: javascript123

Display appropriate messages for:

- Successful login
- Incorrect password
- Invalid username

8. Number Classification System

Write a JavaScript program that accepts an integer from the user and determines whether it is:

- Positive
- Negative
- Zero

Additionally, determine whether it is:

- Even or Odd
- Single-digit, Double-digit, or Multiple-digit

Use nested **if** statements.

9. Admission Eligibility Checker

Write a JavaScript program that accepts marks in:

- Mathematics
- Physics
- Chemistry

A student is eligible for admission if:

- Mathematics ≥ 70
- Physics ≥ 60
- Chemistry ≥ 60
- Total marks ≥ 220

Display appropriate admission status along with reasons if rejected.

10. Age Category using Ternary (?) Operator

Write a JavaScript program that accepts the user's age and displays the category using **nested ternary operators**.

Categories:

- Below 3 → Baby
- 3–12 → Child
- 13–19 → Teenager
- 20–59 → Adult
- 60 and above → Senior Citizen

Display a suitable greeting for each category.

11. Employee Bonus Calculator

Write a JavaScript program that accepts:

- Employee's salary
- Years of experience

Bonus Rules:

- Experience ≥ 10 years → 20% bonus
- Experience ≥ 5 years → 10% bonus
- Otherwise → 5% bonus

Display:

- Bonus amount
- Total salary after bonus

12. Online Shopping Discount System

Write a JavaScript program that accepts the purchase amount and calculates the discount according to the following rules:

- ₹10,000 and above → 25%
- ₹5,000–₹9,999 → 15%
- ₹2,000–₹4,999 → 10%
- Below ₹2,000 → No discount

Display:

- Original amount
- Discount
- Final payable amount

13. Traffic Signal Simulator

Write a JavaScript program that accepts a traffic signal color from the user:

Possible inputs:

- Red
- Yellow
- Green

Display the appropriate action:

- Stop
- Get Ready
- Go

If the input is invalid, display an error message.

14. Banking ATM Menu System

Write a JavaScript program that displays the following menu:

1. Balance Inquiry
2. Cash Withdrawal
3. Cash Deposit
4. Exit

Accept the user's choice and perform the corresponding action using **if-else-if** statements.

For withdrawal:

- Check whether sufficient balance is available.
- Display appropriate messages.

15. Comprehensive User Validation System

Write a JavaScript program that accepts the following information from the user:

- Name
- Age
- Annual Income
- Citizenship
- Qualification

The program should determine:

- Whether the person is eligible for a government job.
- Whether the person can apply for a driving license.
- Whether the person falls under the taxable income category.
- Whether the person is eligible to vote.

Use:

- Nested **if** statements
- **if-else-if** ladder
- At least one **ternary (?) operator**

Display a complete eligibility report for the user.